

1- Which of the following is an example of a system?

- a) Random people on the street
- b) A pile of sand
- c) A car (engine, wheels, transmission working together)
- d) Stones scattered on a beach

2- System definition is not absolute because:

- a) Systems are always the same
- b) It depends on the computer used
- c) It is determined by what questions you're trying to answer
- d) Systems cannot be defined

3- Which of the following is NOT an attribute that characterizes a system?

- a) System boundary
- b) System components
- c) System color
- d) Environment

4- System boundary is defined as:

- a) The physical walls of the system
- b) What separates the system from its environment
- c) The input to the system
- d) The output of the system

5- Environment in a system refers to:

- a) The internal components of the system
- b) The system boundary
- c) Everything outside the boundary that affects the system
- d) The weather only

6- An entity in a system is:

- a) A property of an object
- b) A time period of specified length
- c) An object of interest in the system
- d) A collection of variables

7- An activity in a system represents:

- a) An object of interest
- b) A property of an entity
- c) A time period of specified length
- d) An immediate occurrence

8- The state of a system is defined as:

- a) The physical location of the system
- b) The collection of variables necessary to describe the system at any time
- c) The number of entities in the system
- d) The activities occurring in the system

9- In a bank study, which of the following is NOT a state variable?

- a) Number of busy tellers
- b) Number of customers waiting in line
- c) Customer's name
- d) Arrival time of the next customer

10- An event in a system is:

- a) A time period of specified length
- b) An immediate occurrence that might change the system state
- c) A property of an entity
- d) An object of interest

11- An endogenous event:

- a) Occurs in the environment
- b) Occurs within the system
- c) Cannot be predicted
- d) Always causes system failure

12- A discrete system is one where:

- a) State variables change continuously over time
- b) State variables change only at discrete points in time
- c) No state variables exist
- d) State variables never change

13- Esoteric systems are systems where:

- a) Measurements are easy to obtain
- b) Measurements are quantitative
- c) Measurements are not possible with physical measuring devices
- d) Measurements are always accurate

14- A cascaded system is one where:

- a) Events have no effect on each other
- b) Events mutually affect each other
- c) The effects of events are unilateral (A affects B, B affects C)
- d) Events occur randomly

15- A static simulation model is also called:

- a) Dynamic simulation
- b) Continuous simulation
- c) Monte Carlo simulation
- d) Hybrid simulation

16- A dynamic simulation model:

- a) Represents a system at a particular point in time
- b) Represents systems as they change over time
- c) Has no time component
- d) Cannot be used for banks

17- A deterministic simulation model:

- a) Contains random variables
- b) Contains no random variables
- c) Has probabilistic components
- d) Cannot be verified